

AgriPrime Livestock & E-Commerce System

Software Requirements Specification (SRS)

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Contents

1	Introduction	3
1.1	Project Background	3
1.2	Purpose of Document	3
1.3	Scope	3
2	Client Meeting Case Scenario	3
2.1	Initial Contact	3
2.2	Client Problems Observed	4
2.3	Client Expectations	4
3	Stakeholders	4
4	Functional Requirements	5
4.1	Livestock Management Module	5
4.2	Breeding Management	5
4.3	Veterinary Module	5
4.4	Inventory and Feed Management	6
4.5	HR Management	6
4.6	E-Commerce Module	6
4.7	Order Management System	6
4.8	Logistics Module	7
4.9	Financial Module	7

5	Non Functional Requirements	7
5.1	Performance	7
5.2	Security	7
5.3	Usability	7
5.4	Scalability	7
6	Assumptions and Constraints	8
7	Future Scope	8
8	Conclusion	8

1 Introduction

1.1 Project Background

Agriculture and livestock farming in many regions still rely on manual record keeping. Farmers maintain registers for animal health, breeding cycles, milk production, employee duties, and sales. This traditional system often leads to data loss, incorrect decision making, and financial inefficiency.

The AgriPrime system is proposed as a unified digital platform that manages livestock lifecycle, workforce operations, veterinary health, inventory, logistics, and an online e-commerce marketplace.

1.2 Purpose of Document

This document describes the functional and non-functional requirements gathered from the client after multiple meetings and requirement elicitation sessions.

1.3 Scope

The system will:

- Track livestock from birth/purchase to sale
- Manage veterinary health and vaccination
- Handle employee management and payroll
- Provide online selling platform
- Maintain financial analytics and reporting

2 Client Meeting Case Scenario

2.1 Initial Contact

The client, Mr. Ahmed (Farm Owner), operates a medium-scale livestock farm supplying dairy products and live animals. The meeting took place at the farm office where current manual registers were observed.

2.2 Client Problems Observed

- Animal records stored in notebooks
- Difficulty tracking vaccination schedules
- Employees forgetting assigned duties
- Animals sometimes sold twice due to poor tracking
- No online selling platform

2.3 Client Expectations

The client requested:

- A centralized digital system
- Automatic reminders for animal health
- Easy employee monitoring
- Online marketplace integration
- Profit tracking per animal

3 Stakeholders

- Farm Owner (Primary decision maker)
- Veterinarian
- Farm Workers
- Accountant
- Customers
- Delivery Staff

4 Functional Requirements

4.1 Livestock Management Module

Client Requirement: The client wanted every animal to have a digital identity.

System Solution:

- Unique RFID/Tag ID
- Breed and lineage tracking
- Weight history storage
- Lifecycle status monitoring

User Story: As a Farm Manager, I want to search any animal instantly so that I can view its history.

4.2 Breeding Management

Client emphasized genetic quality improvement.

System provides:

- Heat cycle alerts
- Pregnancy tracking
- Offspring auto-registration

User Story: As a veterinarian, I want breeding alerts so that reproduction success increases.

4.3 Veterinary Module

Client Issue: Vaccinations were often missed.

Solution:

- Automated vaccination reminders
- Treatment logs
- Quarantine tagging

User Story: As a vet, I want medical history instantly accessible.

4.4 Inventory and Feed Management

Client needed feed cost control.

System Features:

- Feed stock monitoring
- Diet planning for animal groups
- Medicine expiry alerts

4.5 HR Management

Client wanted accountability among workers.

System Provides:

- Employee profiles
- Attendance tracking
- Task assignment
- Payroll automation

User Story: As an owner, I want to know which worker completed which task.

4.6 E-Commerce Module

Client Goal: Expand business beyond physical buyers.

Features:

- Online animal listings
- Product marketplace
- Secure checkout
- Customer dashboard

4.7 Order Management System

- Order lifecycle tracking
- Inventory locking
- Status updates

4.8 Logistics Module

Client Concern: Animal delivery safety.

Solution:

- Vehicle tracking
- Route assignment
- Digital gate pass

4.9 Financial Module

Client wanted profit visibility.

System calculates:

- Expenses
- Revenue
- Profit per animal

5 Non Functional Requirements

5.1 Performance

System must handle large animal records efficiently.

5.2 Security

- Role Based Access Control
- JWT Authentication
- Secure payment processing

5.3 Usability

Interface must be simple for non-technical farm staff.

5.4 Scalability

System should support expansion to multiple farms.

6 Assumptions and Constraints

- Internet connectivity available at farm office
- Users have basic smartphone knowledge
- MongoDB used for flexible schema storage

7 Future Scope

- AI disease detection using images
- Smart IoT weight sensors
- Automated milk yield prediction

8 Conclusion

The AgriPrime system transforms traditional livestock management into a smart digital ecosystem. The requirements gathered directly reflect real operational challenges faced by the client and aim to improve efficiency, profitability, and scalability.